

Load Interconnection Request System Impact Study Report

For Queue Project Q12-0105

Salmon Creek 138 kV
October 20, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0105. The Interconnection Customer proposes a 178.9 MW large load (data center) facility interconnecting at the Salmon Creek 138 kV substation in the Pennsylvania area. The request was received on February 13, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 77 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,868,000.

3. Network Impacts

3.1 Load Deliverability Analysis

DEPOE BAY SUBSTATION to SALMON CRK 138 kV: 103.7% loading (181 MVA) following the outage of the FLORENCE MILL SUBSTATION to GADSDEN 138 kV; DFAX = 100.0%.

The SALMON CREEK to GADSDEN SUBSTATION 138 kV L is overloaded to 100.7% of its 183 MVA emergency rating for the loss of the DEPE BAY to SALMON CREEK 138 kV line. The project contributes 100.0% DFAX (178.9 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DEPOE BAY SUBSTATION to SALMON CRK 138 kV	Reconductor line	\$5,692,000	100% (\$5,692,000)
SALMON CREEK to GADSDEN SUBSTATION 138 kV L	Reconductor line	\$1,176,000	100% (\$1,176,000)
	Total allocated to project		\$6,868,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.